Sun LEH / Pythia reproduction — benchmark report

Agent-alone candidates, published prediction values and reference variants calculated in this work.

Agent Scientist: gpt-6.1-sol. Predictor names and units are listed with each measurement.

Published predictions and our predictions

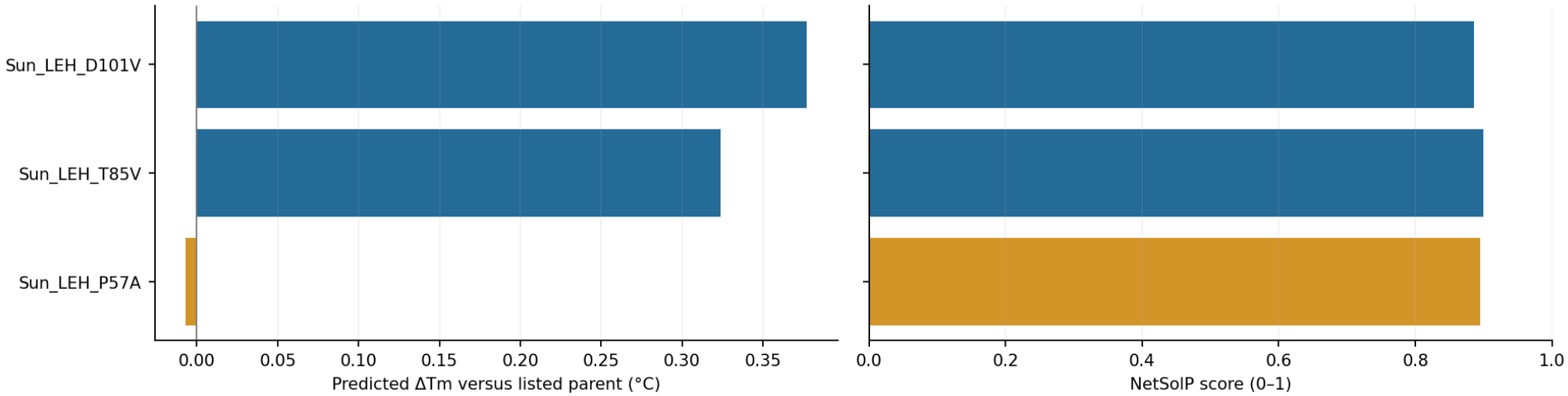
Property / predictor	Published prediction	Our candidates
Pythia score (unitless)	Candidate score not reported	D101V −12.93692160; T85V −7.93473864 (summed score)
Pythia selection cutoff	< −2	33 selected mutations using summed-score implementation
Predicted Tm change (°C)	Not reported	D101V +0.3770; T85V +0.3236

Source: Sun et al., LEH case study

Property predictions calculated in this work

Sequence	Group	Parent / background	Predicted Tm (°C)	ΔTm (°C)	NetSolP	pHPred (raw)
Sun_LEH_D101V	Our candidate	LEH_WT	52.5597	0.3770	0.886254	7.3929
Sun_LEH_T85V	Our candidate	LEH_WT	52.5063	0.3236	0.899697	7.4071
Sun_LEH_P57A	Reference	LEH_WT	52.1758	-0.0069	0.895321	7.3857

Candidate and reference property predictions



Blue: our candidates. Amber: reference variants calculated in this work. Changes use the parents listed in the table.

Parent identity, preparation details and run records: results/task_005_sun_leh_pythia/agent-alone/README.md.